The state of open source in subsurface

97 responses

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About you & your organization

In which region are you located?

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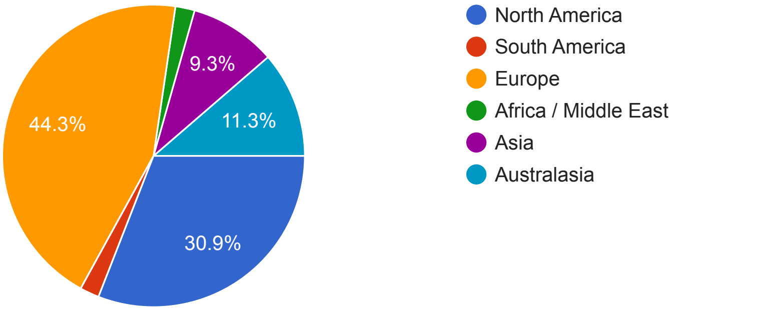
97 responses



What is your primary, current technical domain?

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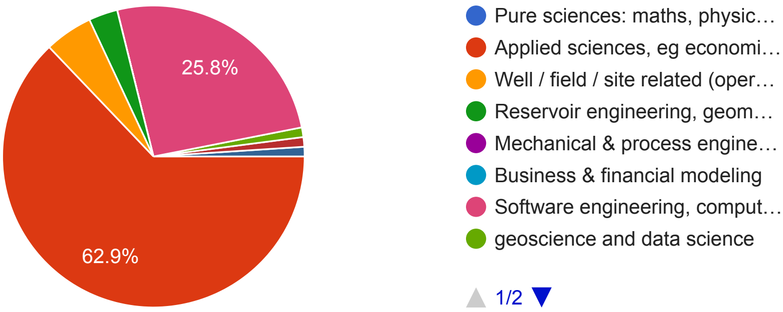
97 responses



During the last 6 months, how much of your work time have you spent writing or reviewing code or documentation?

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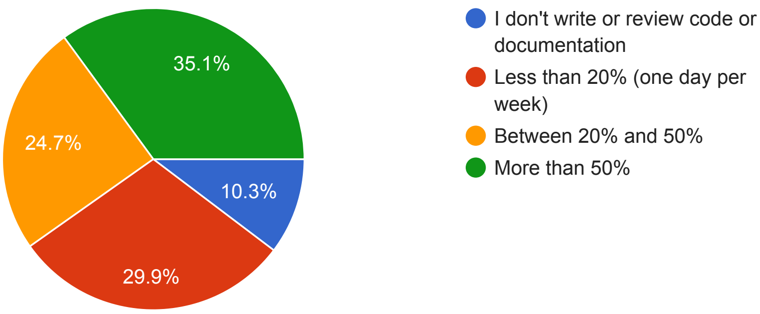
97 responses



Which sector best describes your organization?

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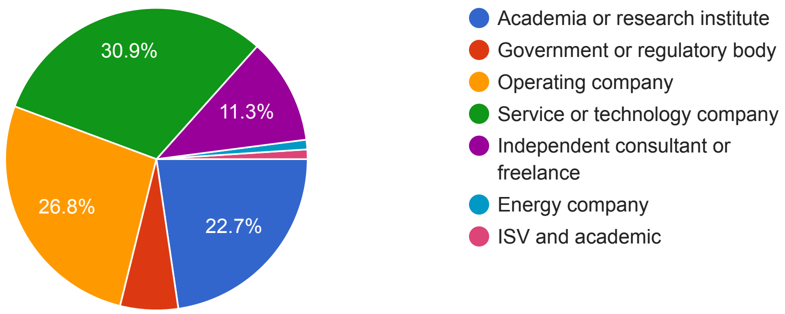
97 responses



What size is your organization?

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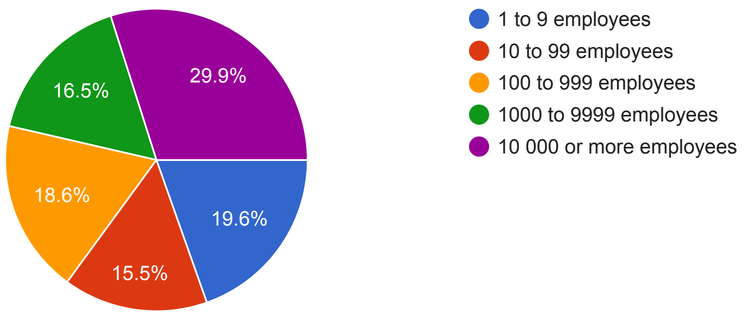
97 responses



Your use of open-source software

When evaluating technical tools for use in a project, how heavily do you and/or your team weigh whether a software option was open-source vs. proprietary?

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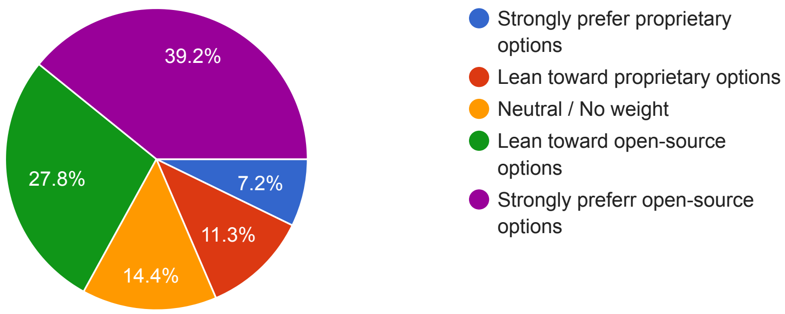
97 responses



In the past 6 months, how often have you **used** open-source software for subsurface or other technical processing, modeling, or analysis?

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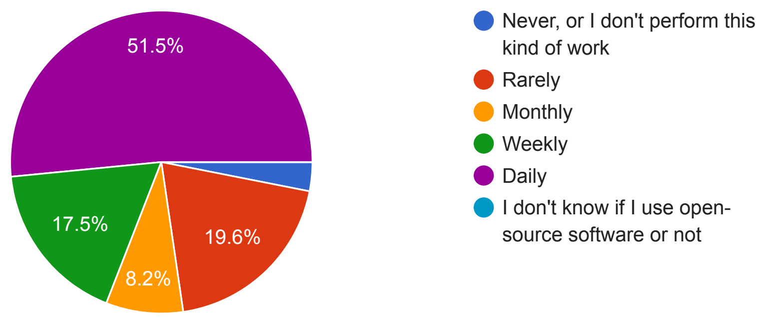
97 responses



Your use of open-source software

Thinking about the open-source subsurface toolset you used in the past 6 months, what were the main reasons for choosing it over a proprietary alternative?

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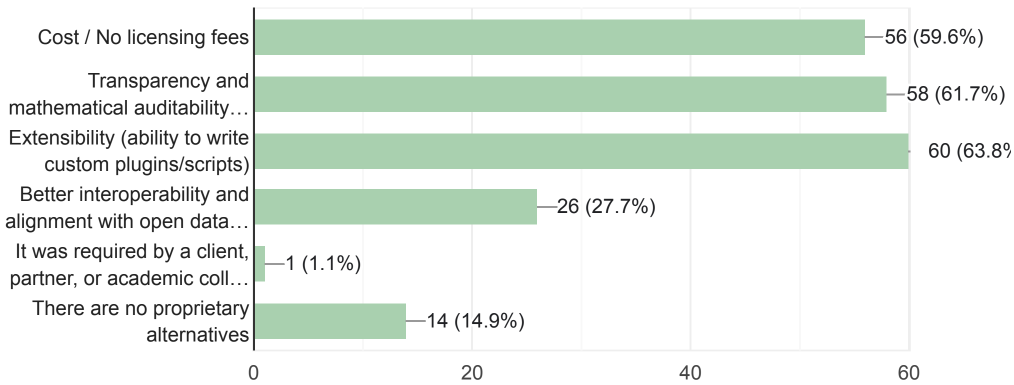
94 responses



In the past 6 months, have you encountered a workflow-breaking bug or security vulnerability in a subsurface open-source tool that delayed your project?

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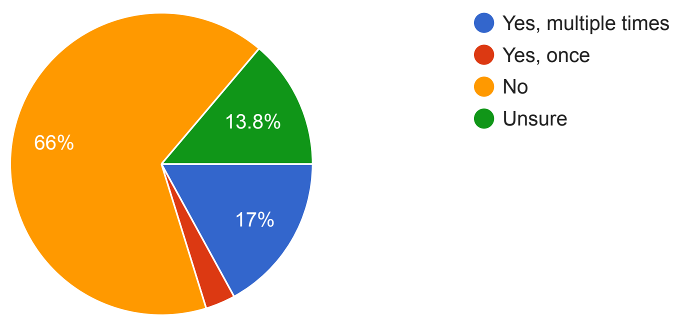
94 responses



Your activities as a core contributor or maintainer

Are you a **core contributor** to any open source repositories? (In any domain, not only subsurface.)

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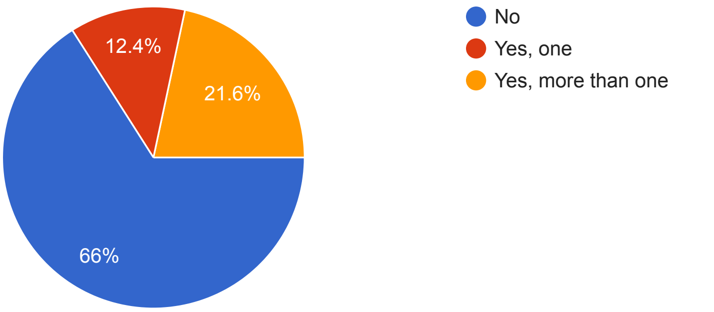
A 'core contributor' is someone with administrative or write access to the repository. This includes the role of 'maintainer'.

97 responses



Your activities as a core contributor or maintainer

Of the repositories to which you are a core contributor, are any of the projects related to subsurface?

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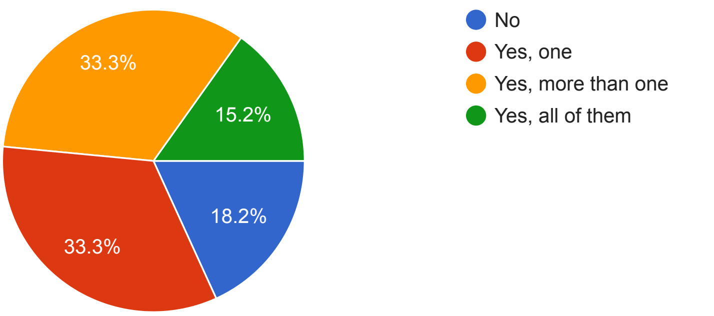
33 responses



Thinking about the primary subsurface open source project you contribute to, how many other core contributors are there on that project?

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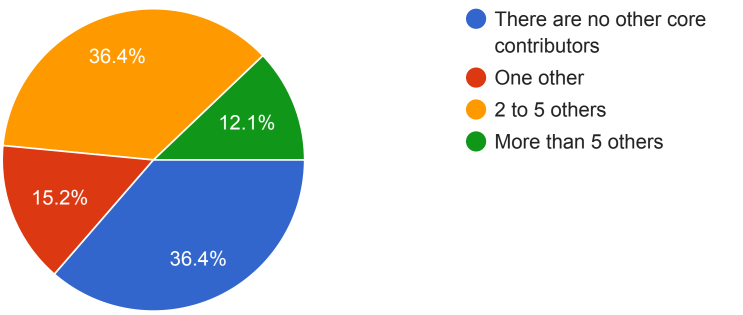
33 responses



Again thinking about this project, how was your time accounted for?

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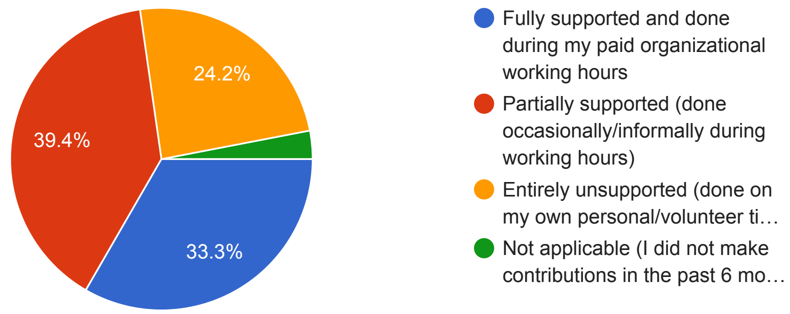
33 responses



Again thinking about this project, what was the primary funding mechanism keeping the subsurface open-source tools you maintain or help maintain afloat?

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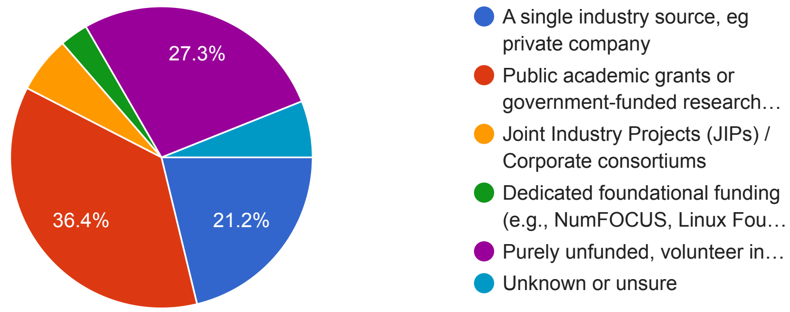
33 responses



Looking ahead to the next 6 to 12 months as a core contributor or maintainer, how do you feel about the following aspects:

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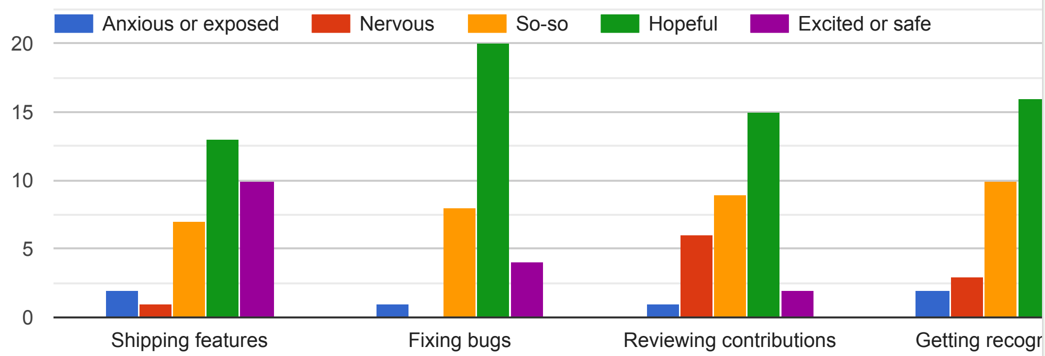
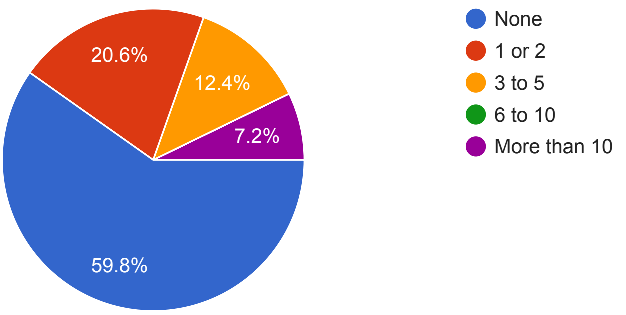


Your activities as a non-core contributor

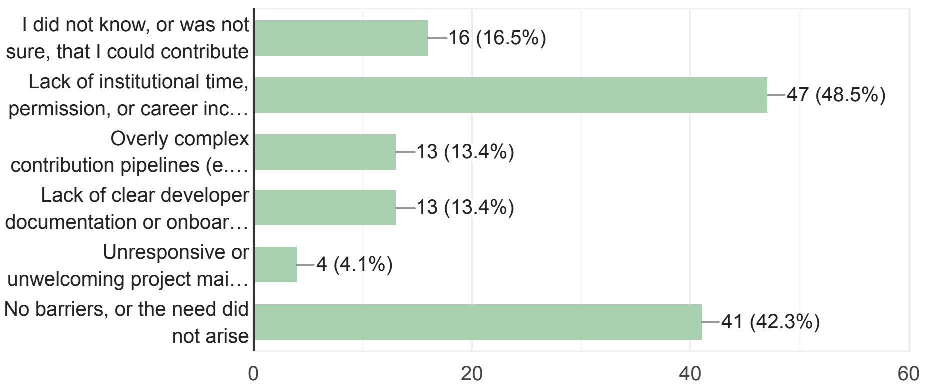
In the past 6 months, how many pull requests (PRs), code commits, or documentation fixes have you submitted to open-source repositories to which you are **not** a core contributor?

97 responses



In the past 6 months, what have been the barriers to contributing code or documentation back to the subsurface tools you use?

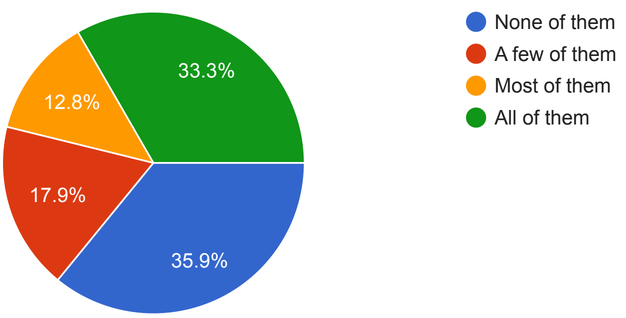
97 responses



Your activities as a non-core contributor

Thinking about your contributions in the last 6 months, what proportion of them were submitted to **subsurface** open-source repositories?

39 responses



In the past 6 months, what kind of contribution(s) have you made? Check all that apply.

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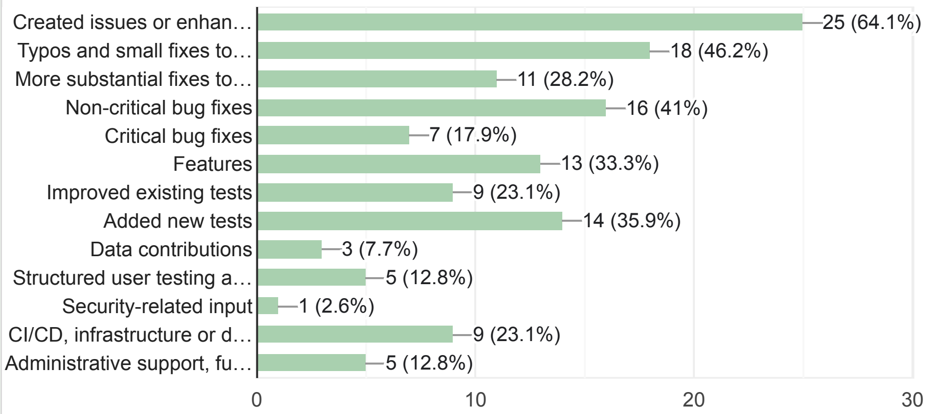
39 responses



Artificial intelligence

How often do you use AI assistants or agentic tools in your work? For example, for help with research, brainstorming, translation, or coding.

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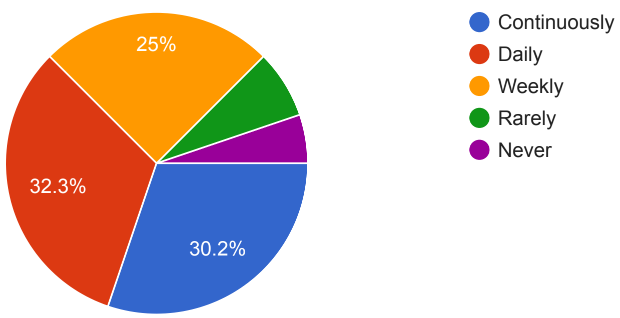
96 responses



To what extent do you trust the assistance of AI assistants and agentic tools?

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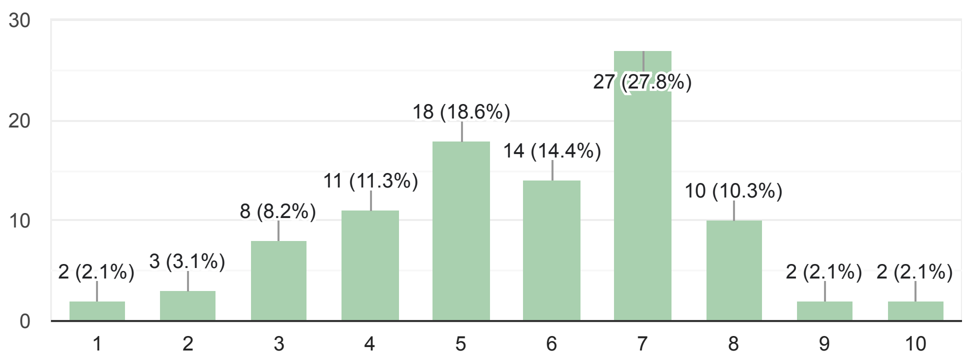
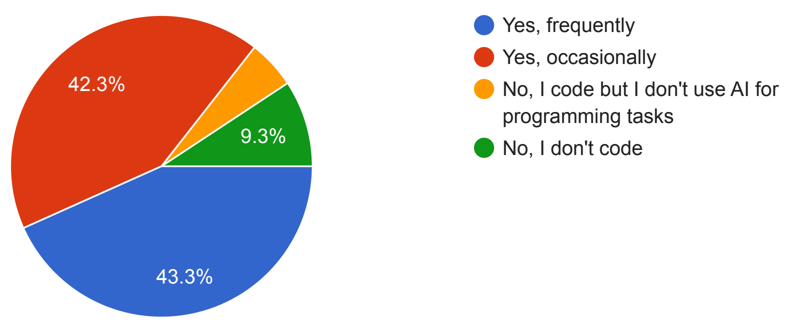
97 responses



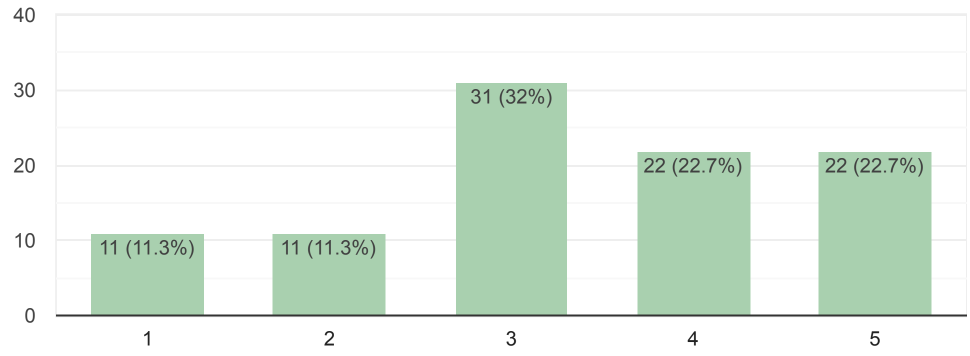
In the past 6 months, have you used AI code assistants (e.g., Copilot, ChatGPT, Claude) to write, debug, or document code meant for subsurface open-source applications?

97 responses



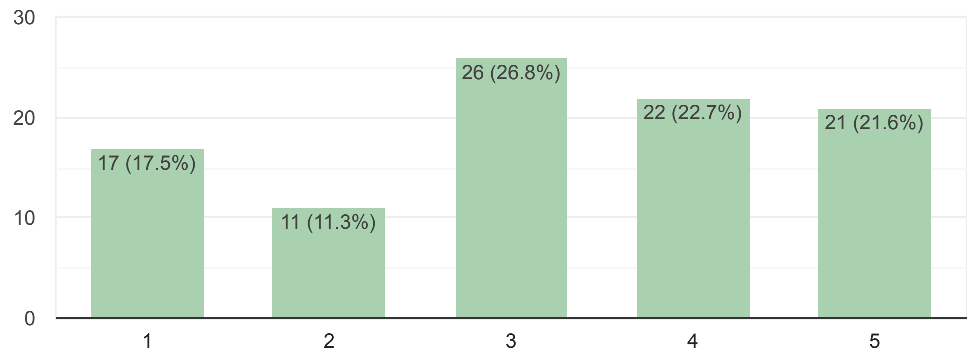
How concerned are you about the security, copyright, or intellectual property (IP) risks of AI-assisted code being merged into open-source repositories your organization **depends on**?

97 responses



How concerned are you about the security, copyright, or intellectual property (IP) risks of AI-assisted code being merged into open-source repositories your organization **produces**?

97 responses



To what extent do you agree with this statement: *"AI code assistants are lowering the barrier for non-programmers to **make use of** subsurface open source projects."*

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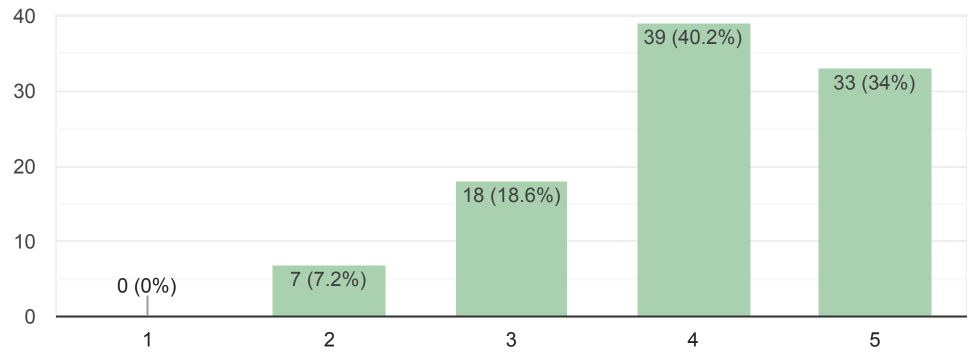
97 responses



To what extent do you agree with this statement: *"AI code assistants are lowering the barrier for non-programmers to **contribute to** subsurface open source projects."*

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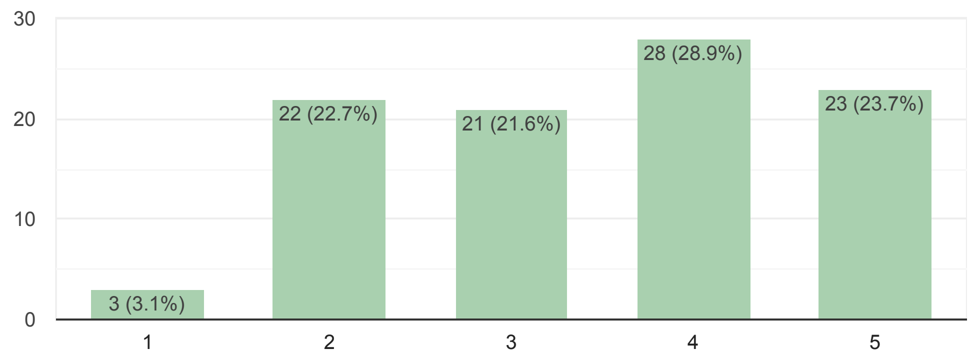
97 responses



Funding and sustainability

In the past 6 months, how would you characterize your organizational leadership's general attitude toward adopting open-source software?

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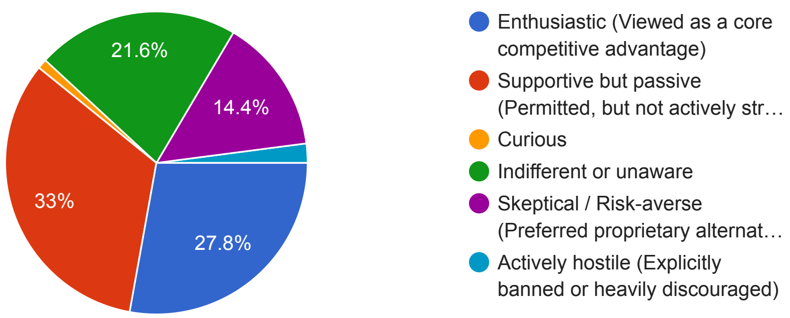
97 responses



Which of the following actions has your organization taken? Select all that apply.

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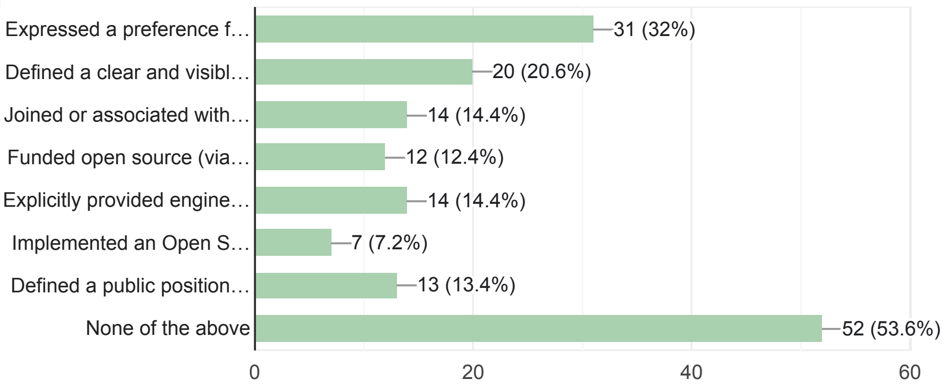
97 responses



In the last 6 months, and to the best of your knowledge, have any of your projects been affected by an upstream dependency doing any of the following:

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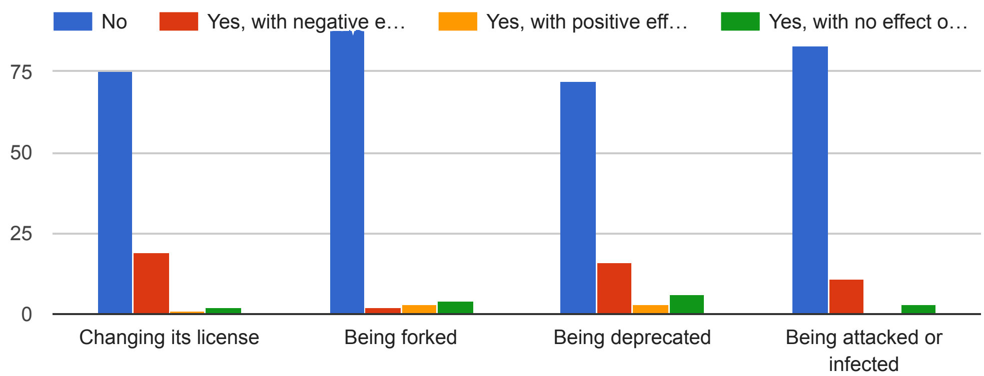

Based on your observations in the past 6 months, how sustainable do you consider the maintenance health of the subsurface open-source ecosystem?

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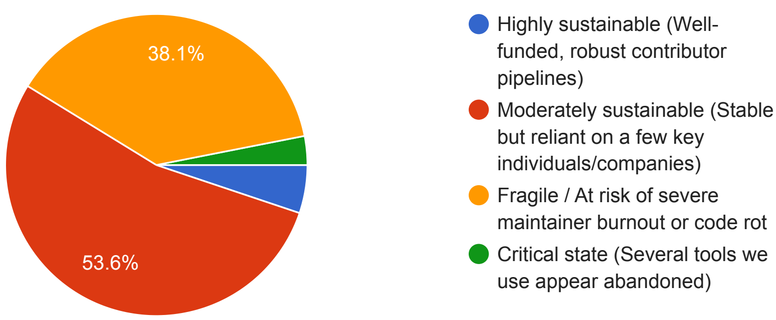
97 responses



That's the end of the survey. Is there anything you'd like to share?

22 responses

Maintaining OSS and securing funding is difficult. Sustainable funding options are urgently needed to keep the momentum.

Thanks for your contributions and sharing of knowledge, time, and enthusiasm.

thank you for doing this

I see an issue of awareness (critical mass). Most if not all opem source software are only known to the few who's genuinely interested. Beyond that it is largely veil from the larger audience who sits behind corporate high walls. More due to ignorance and corporate cordoning than anything else.

I am not concerned with use of AI in open-source workflows. I am concerned with use by people who don't understand the output and use AI for a "one shot" approach to programming.

Working on OpenSource Software builds trusts on the output as compared to closed software

Great initiative!

I feel like there are so many good ideas out there in the geoscience community, and I am enthusiastic about Gen AI being able to surface more of them. They won't be ready for production ideas, but the hope is that, with the help of others, we can mature a variety of ideas and expand our tooling in the open source world.

There is a critical need to support common underlying libraries with goal toward long-term stability, maintainability, and interoperability. One major concern of mine is a trend towards dependency on complex and rapidly changing tools that have short lifetimes, jeopardizing the long-term maintainability of projects. However this work is often hard to sell because it is usually in the background and requires longer timelines compared to more downstream products that address immediately business needs.

The commercial benefit of open source software (OSS) is significant (https://www.hbs.edu/faculty/Pages/item.aspx?num=65230). This has not been well communicated to funding agencies, although it should be a focus point of any OSS communication with the intention of securing greater funding support for OSS.

A key risk to industry of OSS is continuity and on-going support. Mitigating this risk is at the core of developing trusted OSS. e.g. of such trusted OSS are Generic Mapping Tools, pyGIMLi

Next time please also have academicians in mind. I was not able to precisely answer several questions in my capacity as a university professor. Thanks.

AI is reshaping software industry. The way to develop tools and data will be broadly adopted by non-computer science background users.

Similar to AI reducing the barrier to entry for non-programmers to contribute to OSS subsurface projects, it reduces the barrier for entry for non-subsurface-scientists who are programmers. AI's understanding of science is getting better and better. Citations are improving. It's a really incredible time!

Claude, in particular, has opened previously (effectively) inaccessible open-source code to be adopted where before it seemed only the authors could implement. I expect the demise of high-end, commercial 3D geoscience software in only a few years.

My main concern regarding the current state of open-source is how the expectations of almost everyone has shifted to "code and ship fast", especially since LLM tools allow people to quickly generate code that _seems_ like the right solution to the problem.

Even if we have a machine to produce code much faster than before, the work needed to guarantee that such code is well-engineered (has good quality, good design, is reliable, _fails well_, etc.) it's still tied to human capacity. Such capacity requires time to be built, proper learning of foundational knowledge, and mentoring. I see a shift of priorities in our environment: slow processes (learning, mentoring, producing high quality products) now take less attention than shipping features and producing results faster.

Small steps at a big org…

Concerned that the golden age is coming to an end - there's a noticeable reduction in funding and release from large organisations. Equinor is a good example, the period 2017-2023 was a bright moment of things like SEGYIO.

Concern that with the rise of vibe coding, the volume of hack fixes being pushed to unstable untested releases (no dependency checks on wider code modules) is going to push people away from adoption of open source code if there is too much variety, and a lack of guarantee of reliability and maintenance.

Happy Birthday, Matt

Thank you for conducting this survey.

I don't work on sub-surface so some of my questions reflect the wider open-source geospatial community.

While we haven't had dependencies attacked, we now spend significant time updating dependencies where vulnerabilities have been identified.

Good survey

Matt, I just wanted to offer my support! You’re an open-source legend; as a geophysicist, I developed my career inspired by AgileScientific.